LEARN

ARDUINO PROJECTS

UNO R4



BEGINNERS KIT

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Closing Summary

🟦 Introduction to Uno R4 boards and installing Arduino IDE.

About the UNO R4 Board

The UNO R4 is an upgraded version of the Arduino UNO. There are 2 versions of the board available which is Minima and Wifi.

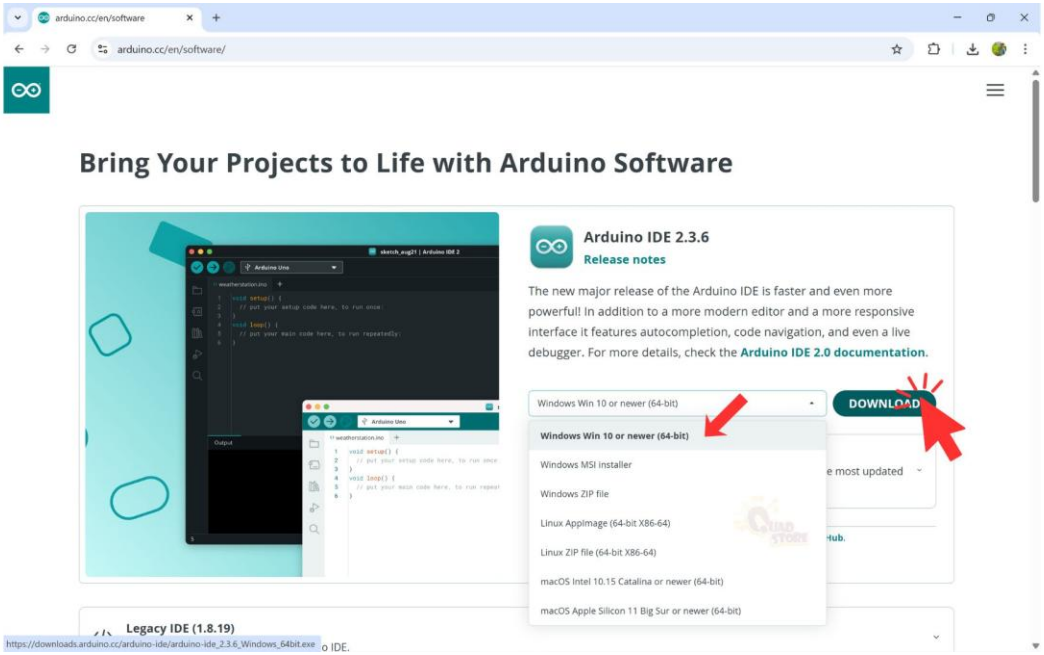
It has more processing power, larger memory, and uses a USB Type-C connector.

Despite improvements, it is still easy to use for beginners.

Installing Arduino IDE

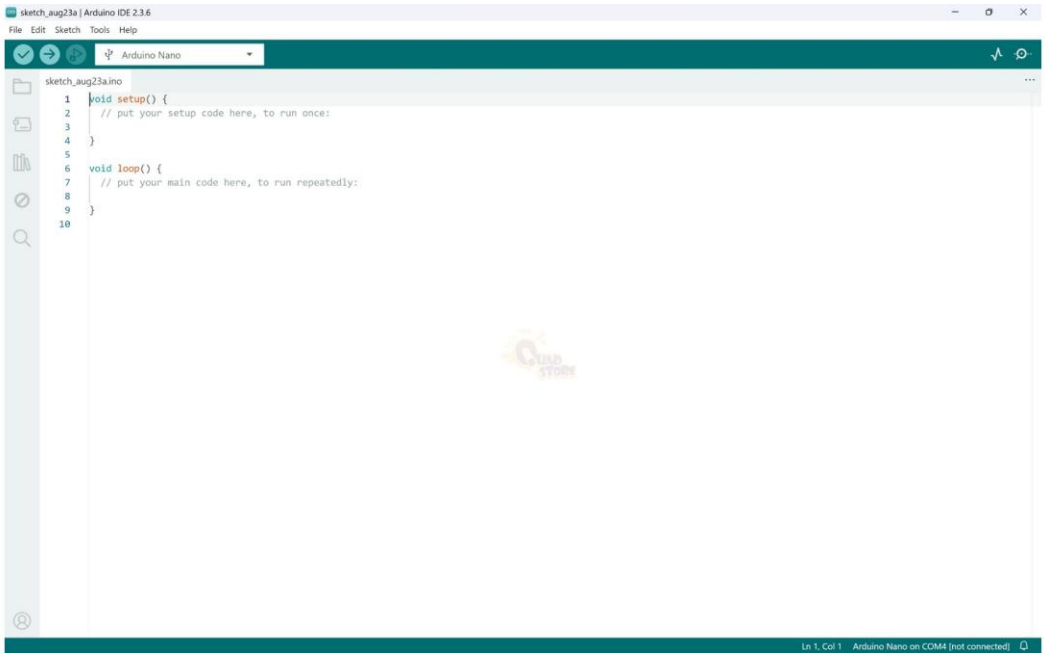
1. Visit the official Arduino website: **https://www.arduino.cc/en/software**

2. Download and install the Arduino IDE for Windows, Mac, or Linux.



3. Open the IDE after installation.

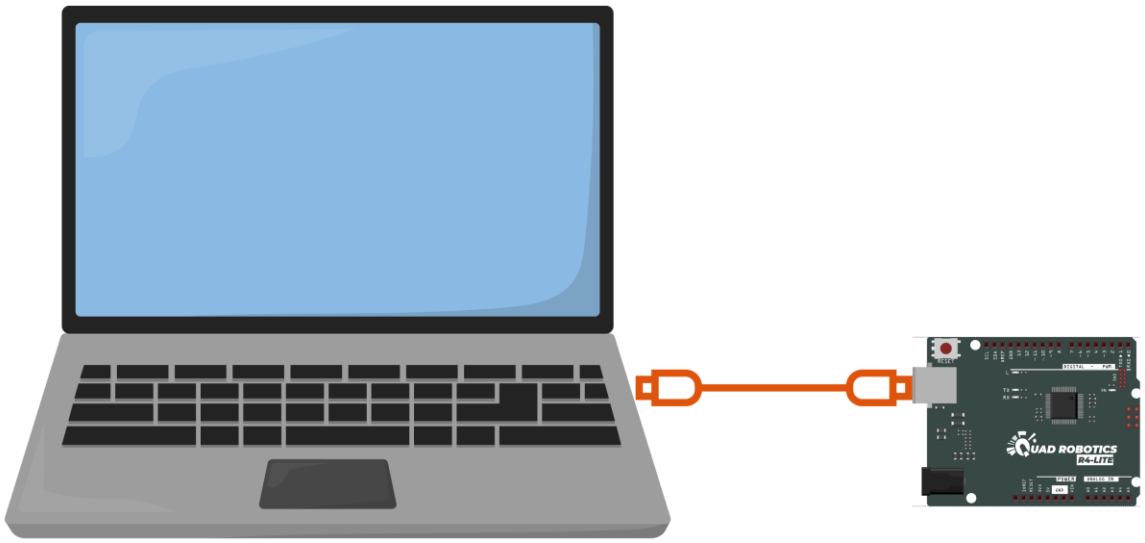




Connecting the UNO R4 Board

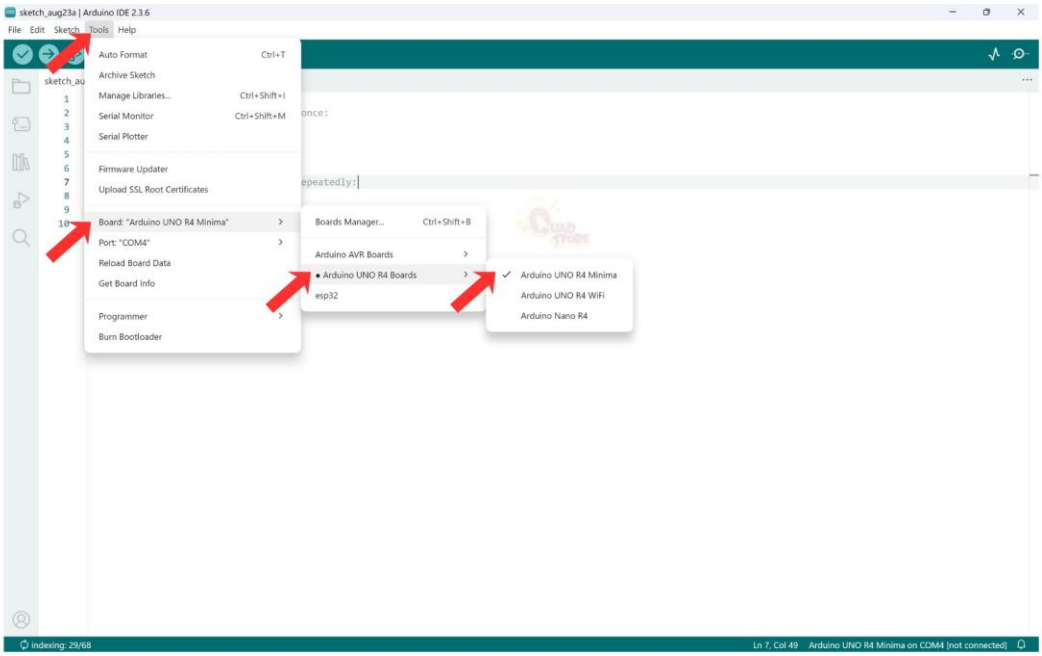
1. Use the Type-C to A USB cable provided in the kit.

2. Plug the Type-C end into the UNO R4 and the USB-A end into your computer.

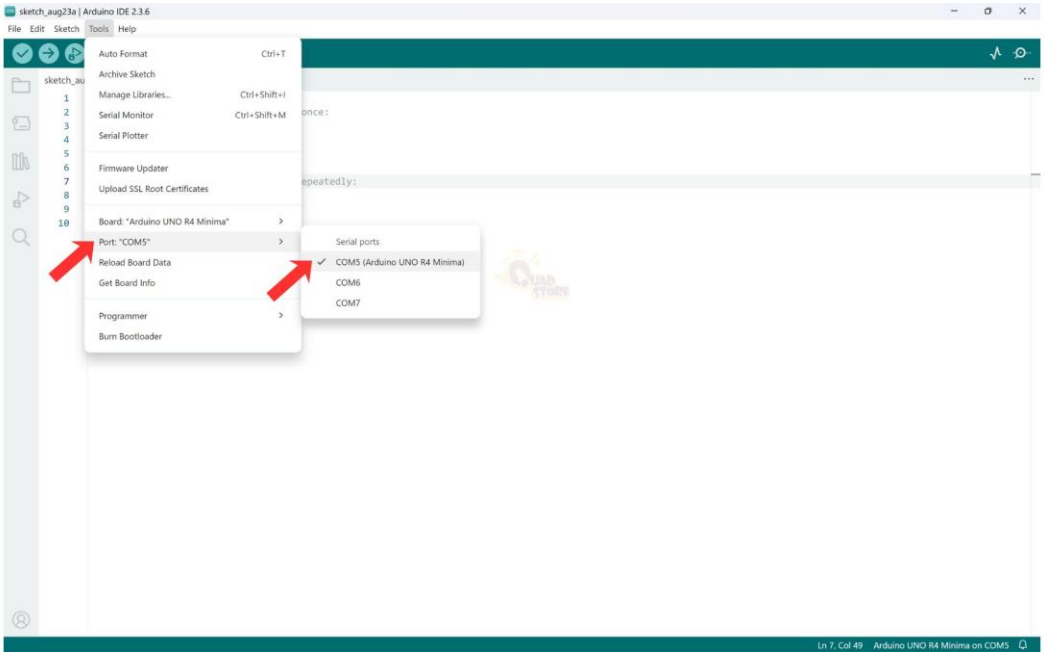


3. The power LED on the board will turn ON.

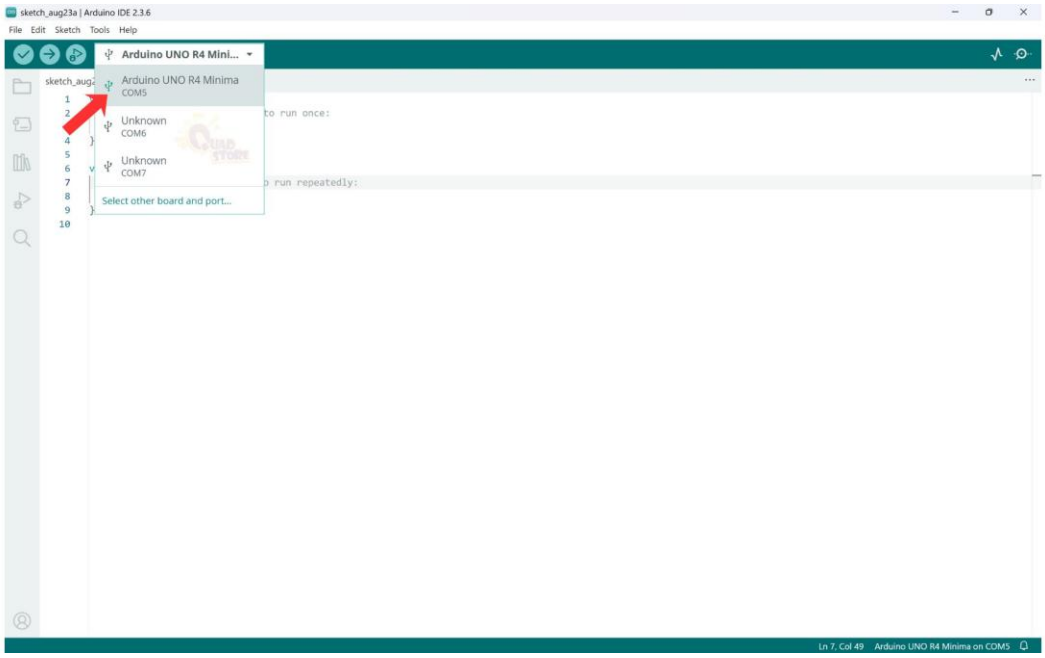
4. Go to **Tools → Board Arduino Uno R4 Board → Select "Arduino Uno R4 Minima" or "Arduino Uno R4 Wifi"** based on your board.



5. Make sure the correct port is selected by going to Tools → Port "COM" → Select the (Arduino Uno R4 boards) based on your COM port. Please note: COM port number will changes based on your computer/laptop.



6. Ensure the correct board is getting reflected in the IDE screen.



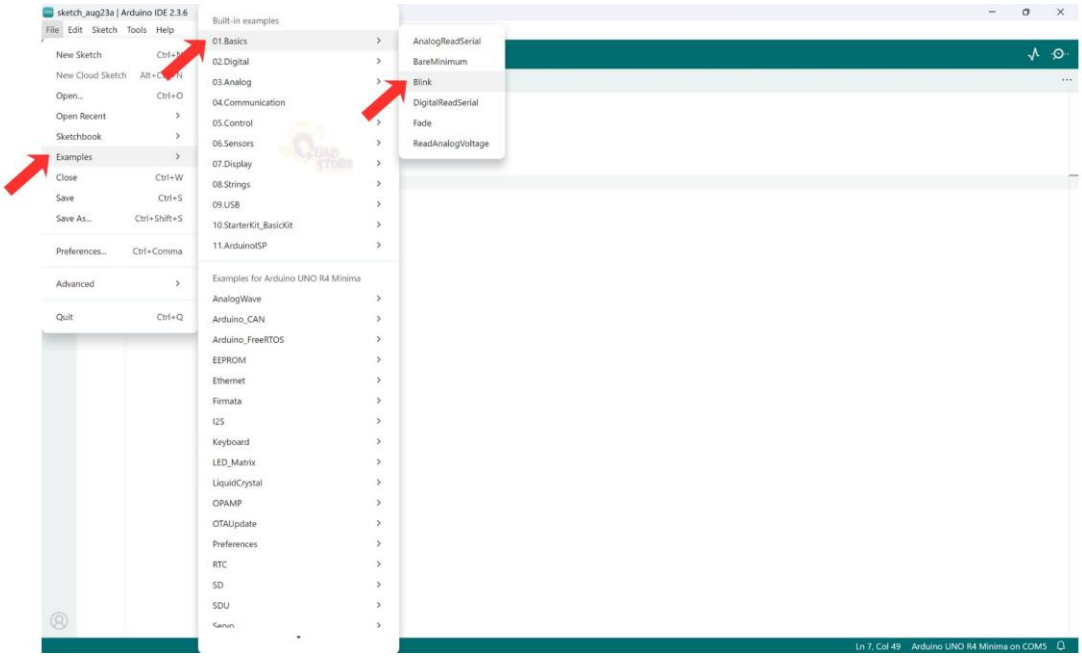
Uploading Your First Program (Blink Test)

This program makes the built-in LED blink and ensures the board and IDE are working correctly.

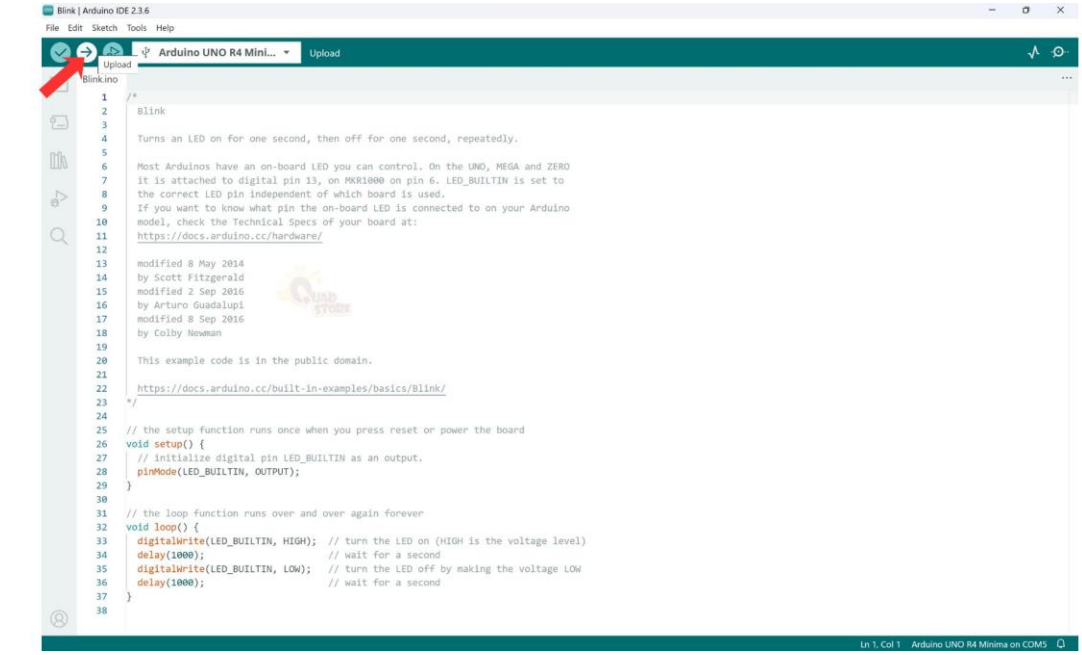
```
void setup() {
  pinMode(LED_BUILTIN, OUTPUT);
}
void loop() {
  digitalWrite(LED_BUILTIN, HIGH);  // Turn LED on
  delay(1000);                      // Wait 1 second
  digitalWrite(LED_BUILTIN, LOW);   // Turn LED off
  delay(1000);                      // Wait 1 second
}
```

Steps to Upload Code

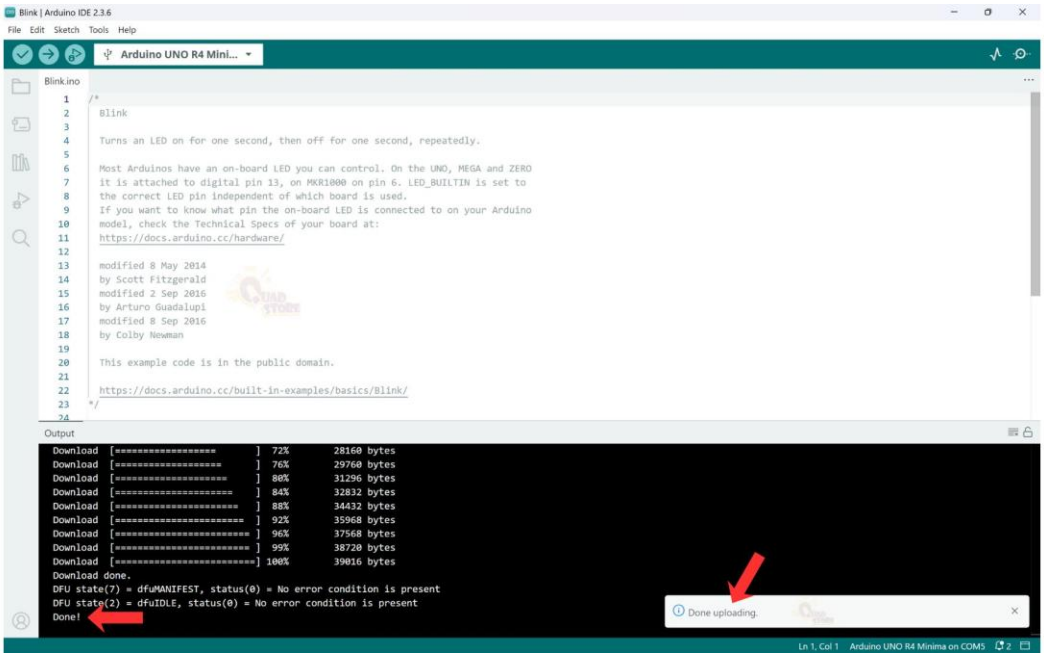
1. Either **copy and paste the above code** into the Arduino IDE (OR) open File → Examples → 0.1 Basics → Blink program..



2. Click on the **Upload button** (right arrow icon).



7. Wait for the code to **compile and upload**. You should see a message "**Done!**" at the bottom of the screen. The board will start running the program automatically.



Output: You should see the built-in LED under PIN 13 starts to blink at a regular interval.

Project 1: External LED Blink

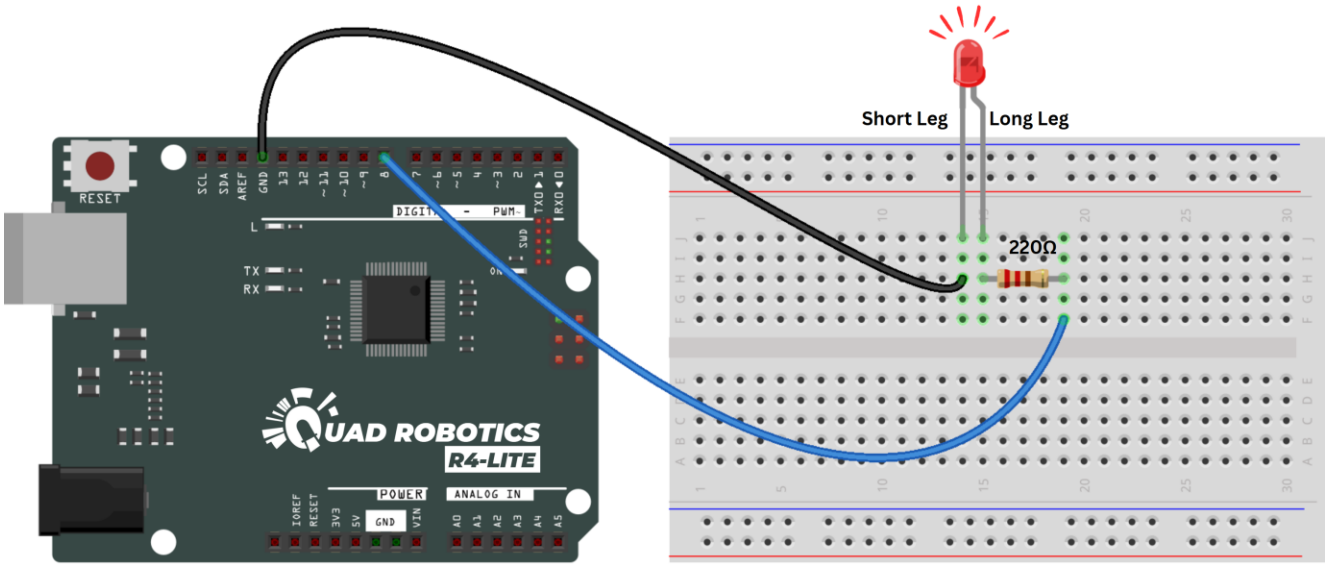
✨ Objective: Blink an external LED connected on a breadboard.

Components Required

- UNO R4 board
- Breadboard
- 1 × LED (any color)
- 1 × 220Ω Resistor
- Jumper wires
- Type C to A USB cable

Circuit Diagram

Connect LED anode (long leg) to pin 8 via 220Ω resistor, and cathode (short leg) to GND.



Code

```
int led = 8;

void setup() {
  pinMode(led, OUTPUT);
}
void loop() {
  digitalWrite(led, HIGH);
  delay(1000);
  digitalWrite(led, LOW);
  delay(1000);
}
```

📥 Steps to Upload Code

1. Connect your **UNO R4 board** to the computer using a **USB cable**.
2. Open **Arduino IDE** on your computer.
3. Go to **Tools > Board** and select **Arduino UNO R4**.
4. Go to **Tools > Port** and choose the correct **COM port**.
5. Either **copy and paste the above code** into the Arduino IDE OR open the file **LEDBlink.ino** from the provided **CODE folder**.
6. Click on the **Upload button** (right arrow icon).
7. Wait for the code to **compile and upload**. The board will start running the program automatically.

🔑 Explanation & Output

An external LED is connected to pin 8 with a 220Ω resistor for safety.

The program turns the LED ON (HIGH) and OFF (LOW) every second.

🛠 DIY Extension

Try different colors of LEDs.

Add more LEDs on pins 9 and 10 for patterns like traffic lights.

Project 2: Traffic Lights with 3 LEDs

✨ Objective: Simulate a simple traffic light system using 3 LEDs.

Components Required

- UNO R4 board

- Breadboard

- 3 × LEDs (Red, Yellow, Green)

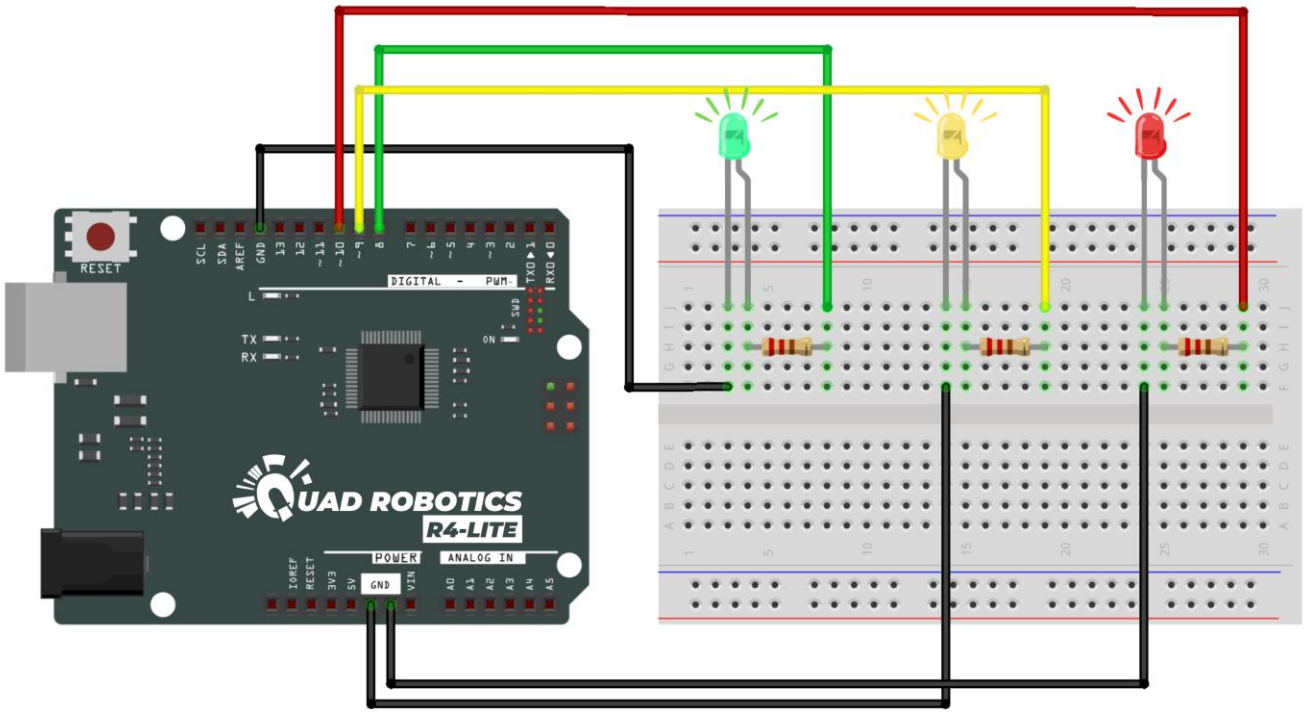
- 3 × 220Ω Resistors

- Jumper wires

- Type C to A USB cable

Circuit Diagram

Red LED → pin 10, Yellow LED → pin 9, Green LED → pin 8 via 220Ω resistors; cathodes to GND.



💻 Code

```
int red = 10;
int yellow = 9;
int green = 8;

void setup() {
  pinMode(red, OUTPUT);
  pinMode(yellow, OUTPUT);
  pinMode(green, OUTPUT);
}
void loop() {
  digitalWrite(red, HIGH);
  delay(3000);
  digitalWrite(red, LOW);

  digitalWrite(yellow, HIGH);
  delay(1000);
  digitalWrite(yellow, LOW);

  digitalWrite(green, HIGH);
  delay(3000);
  digitalWrite(green, LOW);
}
```

📥 Steps to Upload Code

1. Connect your **UNO R4 board** to the computer using a **USB cable**.
2. Open **Arduino IDE** on your computer.
3. Go to **Tools > Board** and select **Arduino UNO R4**.
4. Go to **Tools > Port** and choose the correct **COM port**.
5. Either **copy and paste the above code** into the Arduino IDE OR open the file **TrafficLights.ino** from the provided **CODE folder**.
6. Click on the **Upload button** (right arrow icon).
7. Wait for the code to **compile and upload**. The board will start running the program automatically.

🔑 Explanation & Output

This program simulates a traffic light cycle.

Red stays ON for 3 seconds, Yellow for 1 second, and Green for 3 seconds.

🛠️ DIY Extension

Add a pedestrian button to change the lights when pressed.

Use a buzzer for sound alerts.

Project 3: Passive Buzzer Sound

✨ Objective: Generate sound using a passive buzzer with tone function.

Components Required

- UNO R4 board

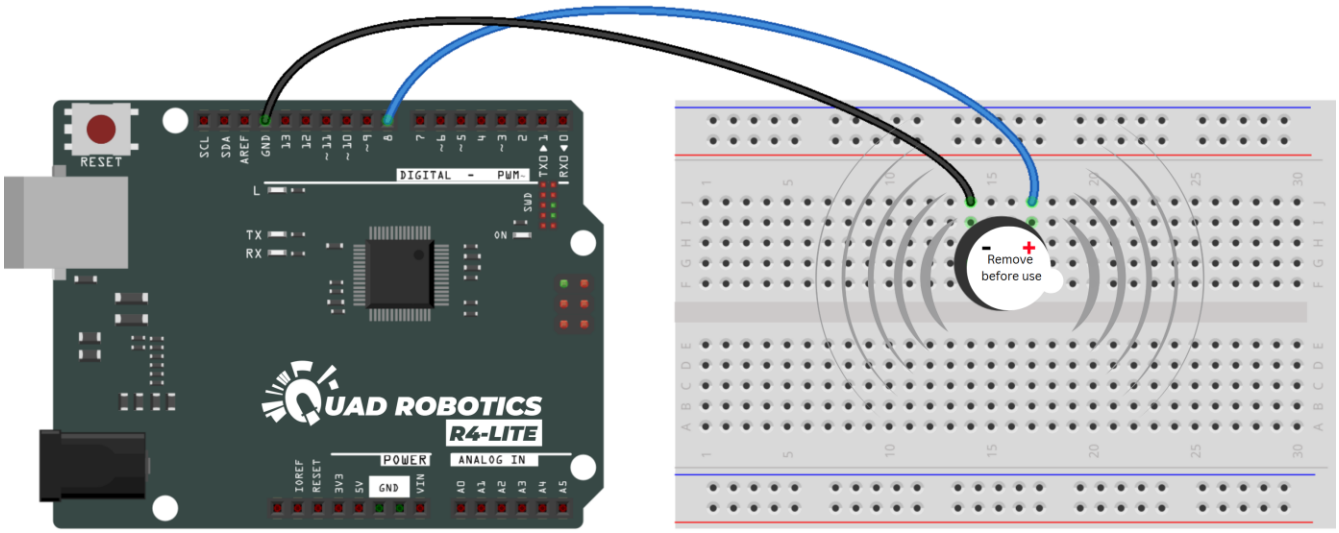
- Passive Buzzer

- Jumper wires

- Breadboard

Circuit Diagram

Connect buzzer + to pin 8, - to GND.



Code

```
int buzzer = 8;

void setup() {
}

void loop() {
  tone(buzzer, 1000); // 1kHz tone
  delay(1000);
  noTone(buzzer);
  delay(1000);
}
```

📥 Steps to Upload Code

1. Connect your **UNO R4 board** to the computer using a **USB cable**.
2. Open **Arduino IDE** on your computer.
3. Go to **Tools > Board** and select **Arduino UNO R4**.
4. Go to **Tools > Port** and choose the correct **COM port**.
5. Either **copy and paste the above code** into the Arduino IDE OR open the file **PassiveBuzzer.ino** from the provided **CODE folder**.
6. Click on the **Upload button** (right arrow icon).
7. Wait for the code to **compile and upload**. The board will start running the program automatically.

🔑 Explanation & Output

The passive buzzer produces sound when driven with different frequencies.

Here we generate a 1kHz tone for 1 second ON and 1 second OFF.

🛠 DIY Extension

Change frequency for different musical notes.

Try making a melody.

Project 4: Active Buzzer Beep

✨ Objective: Turn an active buzzer ON and OFF.

Components Required

- UNO R4 board

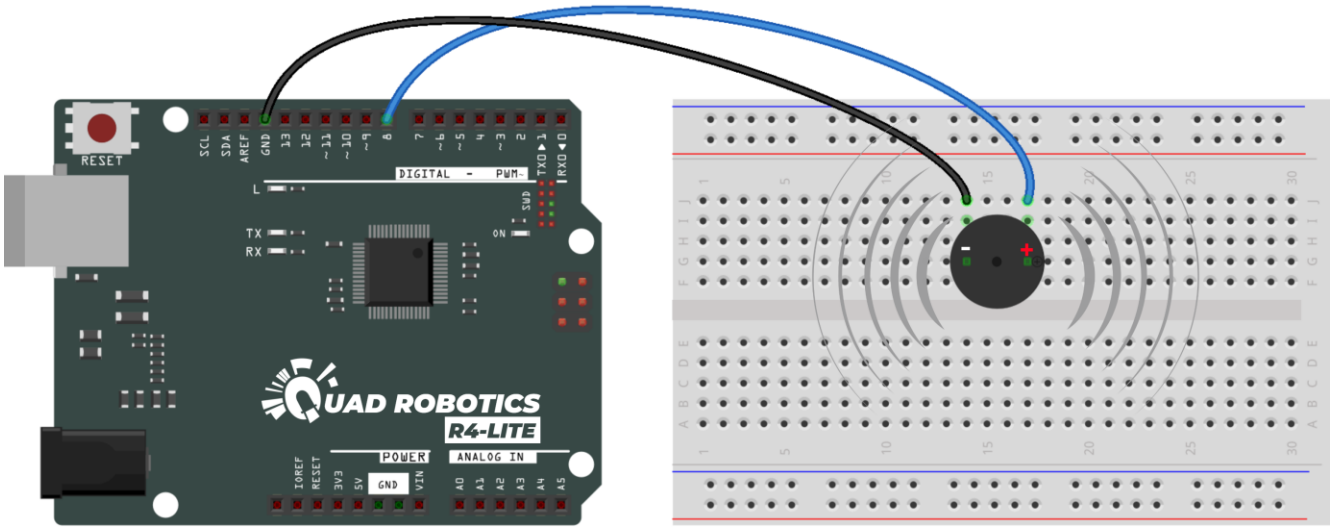
- Active Buzzer

- Jumper wires

- Breadboard

Circuit Diagram

Connect buzzer + to pin 8, - to GND.



Code

```
int buzzer = 8;

void setup() {
  pinMode(buzzer, OUTPUT);
}

void loop() {
  digitalWrite(buzzer, HIGH);
  delay(1000);
  digitalWrite(buzzer, LOW);
  delay(1000);
}
```

📥 Steps to Upload Code

1. Connect your **UNO R4 board** to the computer using a **USB cable**.
2. Open **Arduino IDE** on your computer.
3. Go to **Tools > Board** and select **Arduino UNO R4**.
4. Go to **Tools > Port** and choose the correct **COM port**.
5. Either **copy and paste the above code** into the Arduino IDE OR open the file **ActiveBuzzer.ino** from the provided **CODE folder**.
6. Click on the **Upload button** (right arrow icon).
7. Wait for the code to **compile and upload**. The board will start running the program automatically.

🔑 Explanation & Output

An active buzzer produces a fixed beep when powered.

We simply switch it ON and OFF every second.

🛠 DIY Extension

Use buzzer for alarms in sensor projects.

Project 5: RGB LED Color Mixing

✨ Objective: Control an RGB LED to show different colors.

Components Required

- UNO R4 board

- RGB LED

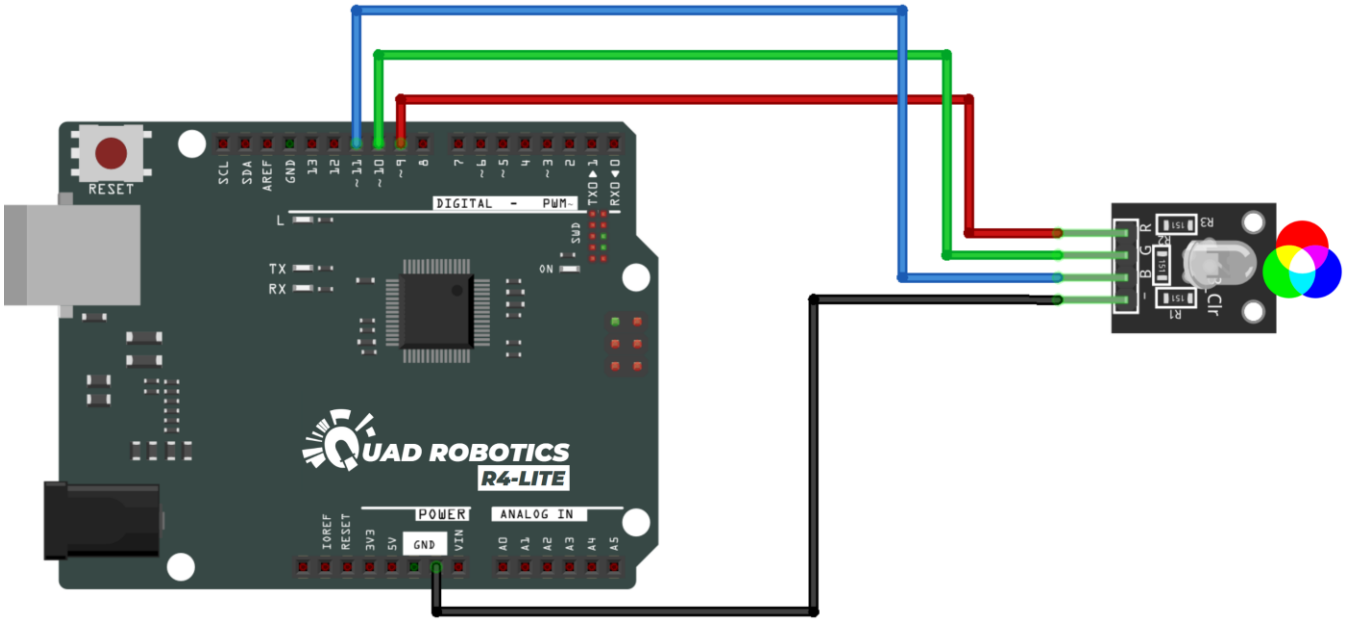
- 3 × 220Ω Resistors

- Breadboard

- Jumper wires

Circuit Diagram

Connect R → pin 9, G → pin 10, B → pin 11 each via 220Ω resistor; common cathode to GND.



💻 Code

```
int red = 9;
int green = 10;
int blue = 11;

void setup() {
  pinMode(red, OUTPUT);
  pinMode(green, OUTPUT);
  pinMode(blue, OUTPUT);
}

void loop() {
  digitalWrite(red, HIGH);
  digitalWrite(green, LOW);
  digitalWrite(blue, LOW);
  delay(1000);

  digitalWrite(red, LOW);
  digitalWrite(green, HIGH);
  delay(1000);

  digitalWrite(green, LOW);
  digitalWrite(blue, HIGH);
  delay(1000);
}
```

📥 Steps to Upload Code

1. Connect your **UNO R4 board** to the computer using a **USB cable**.
2. Open **Arduino IDE** on your computer.
3. Go to **Tools > Board** and select **Arduino UNO R4**.
4. Go to **Tools > Port** and choose the correct **COM port**.
5. Either **copy and paste the above code** into the Arduino IDE OR open the file **RGBLED.ino** from the provided **CODE folder**.
6. Click on the **Upload button** (right arrow icon).
7. Wait for the code to **compile and upload**. The board will start running the program automatically.

🔑 Explanation & Output

An RGB LED combines red, green, and blue light to produce colors.
We switch between red, green, and blue every second.

🛠️ DIY Extension

Mix multiple colors by turning ON two pins together.
Use analogWrite for smooth fading.

Project 6: Potentiometer-controlled LED Brightness

✨ Objective: Use a potentiometer to adjust LED brightness (PWM).

Components Required

- UNO R4 board

- 10K Potentiometer

- LED

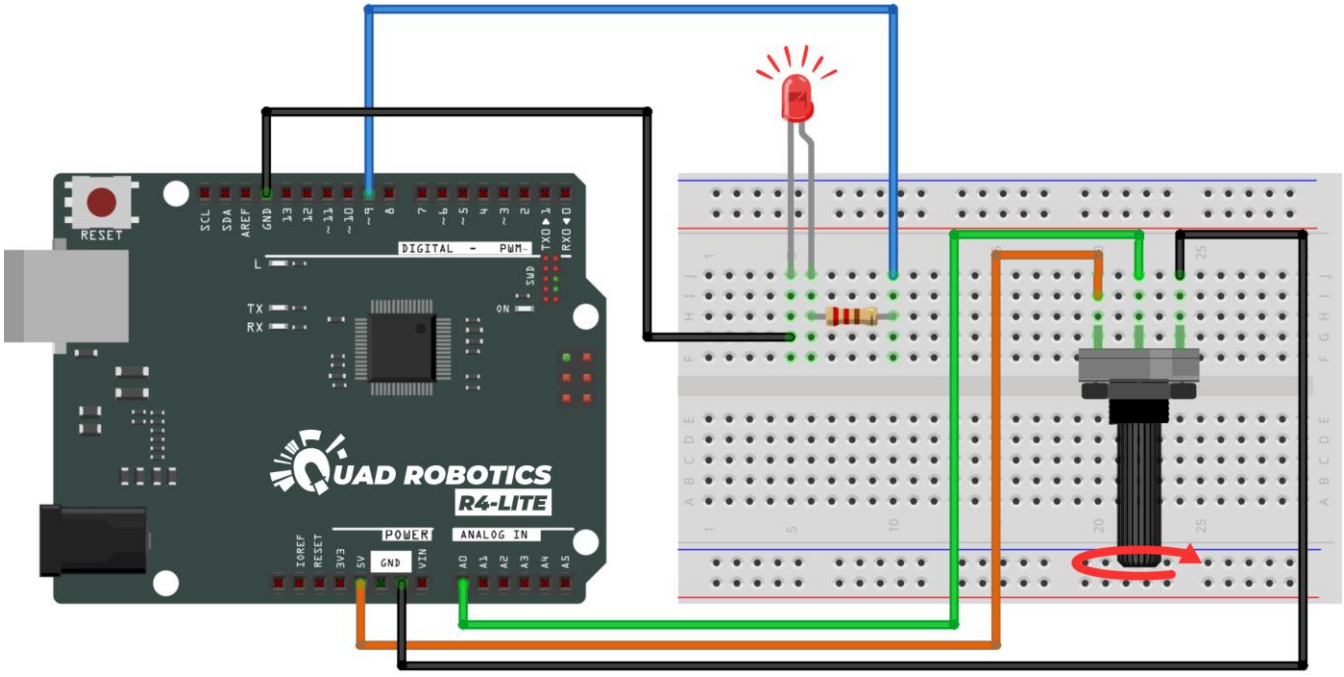
- 220Ω Resistor

- Breadboard

- Jumper wires

Circuit Diagram

Potentiometer middle pin → A0, Left side → 5V and Right side → GND.
LED → pin 9 via resistor.



💻 Code

```
int potPin = A0;
int led = 9;
int val = 0;

void setup() {
  pinMode(led, OUTPUT);
}

void loop() {
  val = analogRead(potPin);
  val = map(val, 0, 1023, 0, 255);
  analogWrite(led, val);
}
```

📥 Steps to Upload Code

1. Connect your **UNO R4 board** to the computer using a **USB cable**.
2. Open **Arduino IDE** on your computer.
3. Go to **Tools > Board** and select **Arduino UNO R4**.
4. Go to **Tools > Port** and choose the correct **COM port**.
5. Either **copy and paste the above code** into the Arduino IDE OR open the file **PotLED.ino** from the provided **CODE folder**.
6. Click on the **Upload button** (right arrow icon).
7. Wait for the code to **compile and upload**. The board will start running the program automatically.

🔑 Explanation & Output

Potentiometer provides analog input from 0–1023.

We map it to 0–255 to control LED brightness with PWM.

🛠️ DIY Extension

Try controlling buzzer pitch with potentiometer.

Project 7: LDR Night Lamp

✨ Objective: Turn ON an LED in darkness using LDR sensor.

Components Required

- UNO R4 board

- LDR Module

- LED

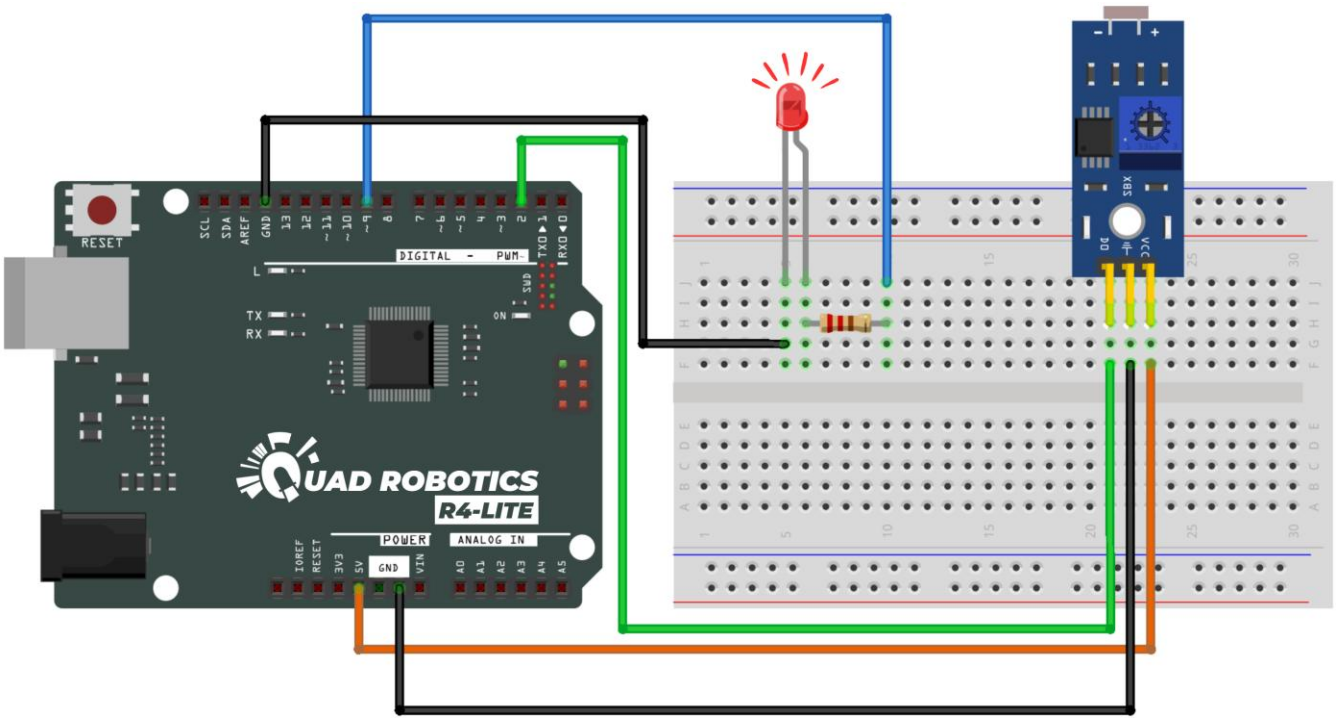
- 220Ω Resistor

- Breadboard

- Jumper wires

Circuit Diagram

LDR output → pin 2,
LED → pin 9 via resistor.



💻 Code

```
int ldr = 2;
int led = 9;

void setup() {
      pinMode(led, OUTPUT);

      pinMode(ldr, INPUT);
}

void loop() {
  int val = digitalRead(ldr);
  if(val == 500){
    digitalWrite(led, HIGH);
  } else {
    digitalWrite(led, LOW);
  }
}
```

📥 Steps to Upload Code

1. Connect your **UNO R4 board** to the computer using a **USB cable**.
2. Open **Arduino IDE** on your computer.
3. Go to **Tools > Board** and select **Arduino UNO R4**.
4. Go to **Tools > Port** and choose the correct **COM port**.
5. Either **copy and paste the above code** into the Arduino IDE OR open the file **LDRLamp.ino** from the provided **CODE folder**.
6. Click on the **Upload button** (right arrow icon).
7. Wait for the code to **compile and upload**. The board will start running the program automatically.

🔑 Explanation & Output

The LDR detects light level.

LED turns ON when it's dark (low value).

🛠 DIY Extension

Adjust threshold value for sensitivity.

Use for automatic room lights.

Project 8: Tilt Sensor Alert

✨ Objective: Detect tilt movement using SW-200D sensor.

Components Required

- UNO R4 board

- Tilt Sensor

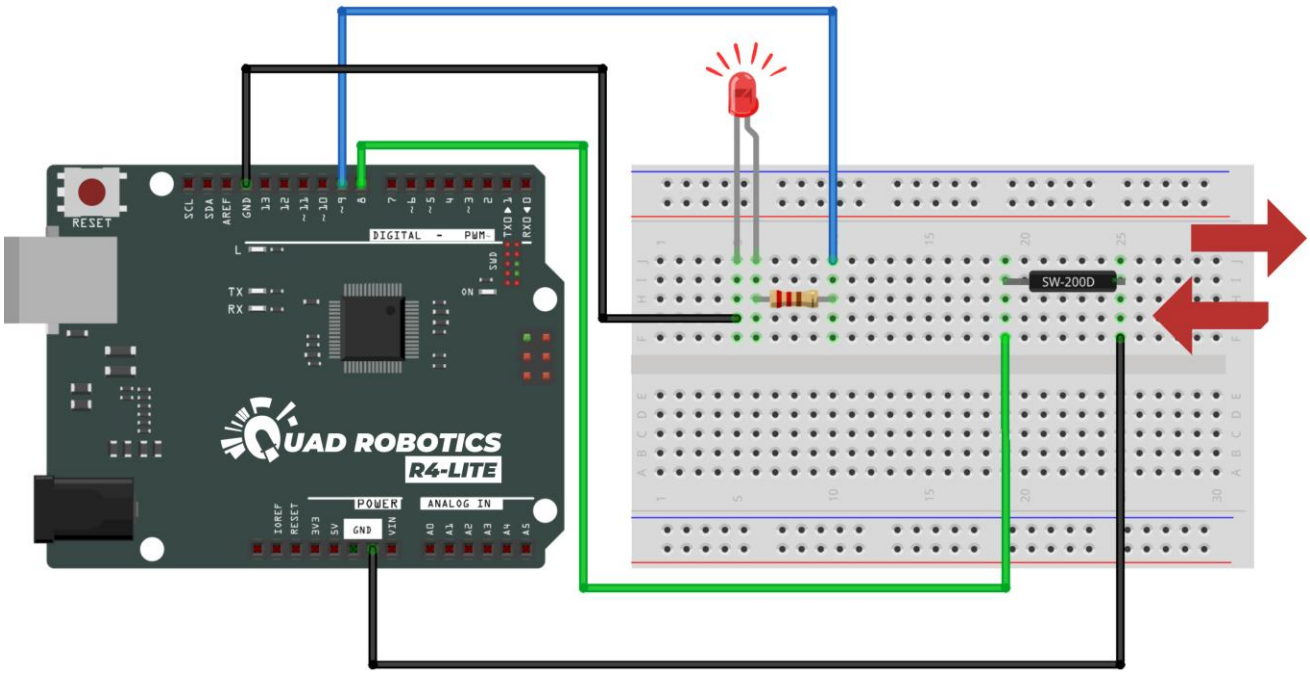
- Buzzer or LED

- Breadboard

- Jumper wires

Circuit Diagram

Tilt sensor output → pin 8, other pin → GND.
LED/buzzer → pin 9.



💻 Code

```
int tilt = 8;
int led = 9;

void setup() {
  pinMode(tilt, INPUT);
  pinMode(led, OUTPUT);
}

void loop() {
  if(digitalRead(tilt) == HIGH){
    digitalWrite(led, HIGH);
  } else {
    digitalWrite(led, LOW);
  }
}
```

📥 Steps to Upload Code

1. Connect your **UNO R4 board** to the computer using a **USB cable**.
2. Open **Arduino IDE** on your computer.
3. Go to **Tools > Board** and select **Arduino UNO R4**.
4. Go to **Tools > Port** and choose the correct **COM port**.
5. Either **copy and paste the above code** into the Arduino IDE OR open the file **TiltSensor.ino** from the provided **CODE folder**.
6. Click on the **Upload button** (right arrow icon).
7. Wait for the code to **compile and upload**. The board will start running the program automatically.

🔑 Explanation & Output

Tilt sensor works like a switch when moved.

Take the breadboard and tilt it in either direction to see the LED/buzzer turn on and off when tilted.

🛠 DIY Extension

Use in anti-theft alarms.

Project 9: Servo Motor Sweep

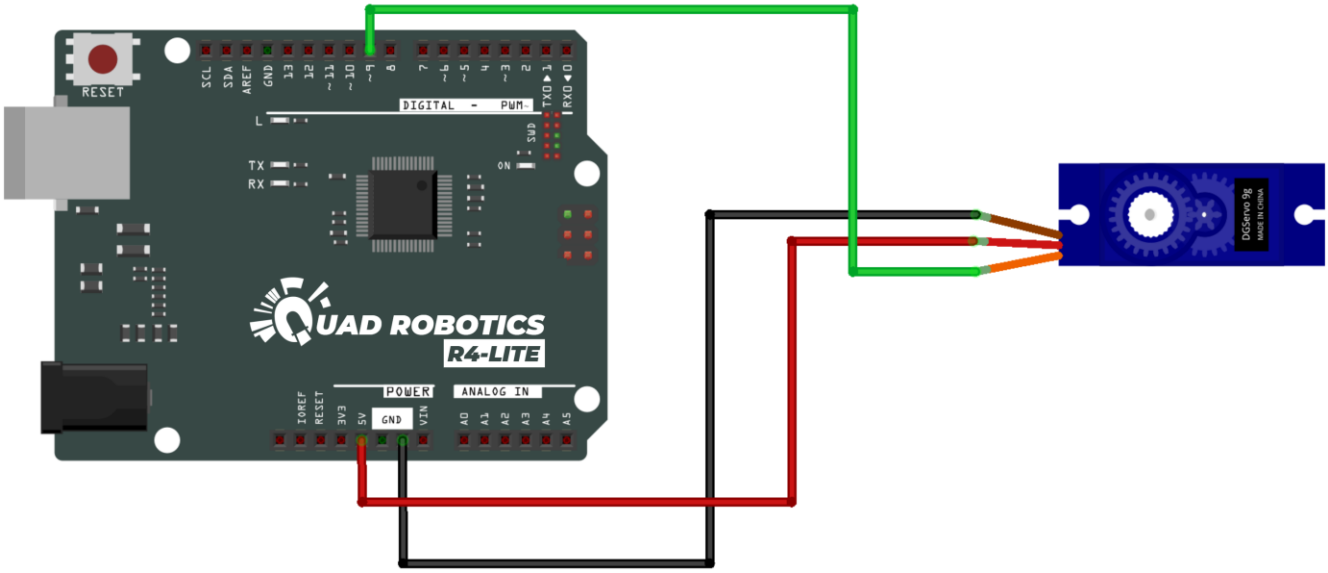
✨ Objective: Move SG90 servo back and forth.

Components Required

- UNO R4 board
- SG90 Servo
- Jumper wires

Circuit Diagram

Servo signal → pin 9, VCC → 5V, GND → GND.



Code

```
#include <Servo.h>

Servo myservo;

void setup() {
  myservo.attach(9);
}

void loop() {
  for(int pos=0; pos<=180; pos++){
    myservo.write(pos);
    delay(15);
  }
  for(int pos=180; pos>=0; pos--){
    myservo.write(pos);
    delay(15);
  }
}
```

📥 Steps to Upload Code

1. Connect your **UNO R4 board** to the computer using a **USB cable**.
2. Open **Arduino IDE** on your computer.
3. Go to **Tools > Board** and select **Arduino UNO R4**.
4. Go to **Tools > Port** and choose the correct **COM port**.
5. Either **copy and paste the above code** into the Arduino IDE OR open the file **ServoSweep.ino** from the provided **CODE folder**.
6. Click on the **Upload button** (right arrow icon).
7. Wait for the code to **compile and upload**. The board will start running the program automatically.

🔑 Explanation & Output

Servo is controlled by PWM signals.

We sweep from 0° to 180° and back.

🛠 DIY Extension

Use for robot arms.

Project 10: Ultrasonic Distance Finder

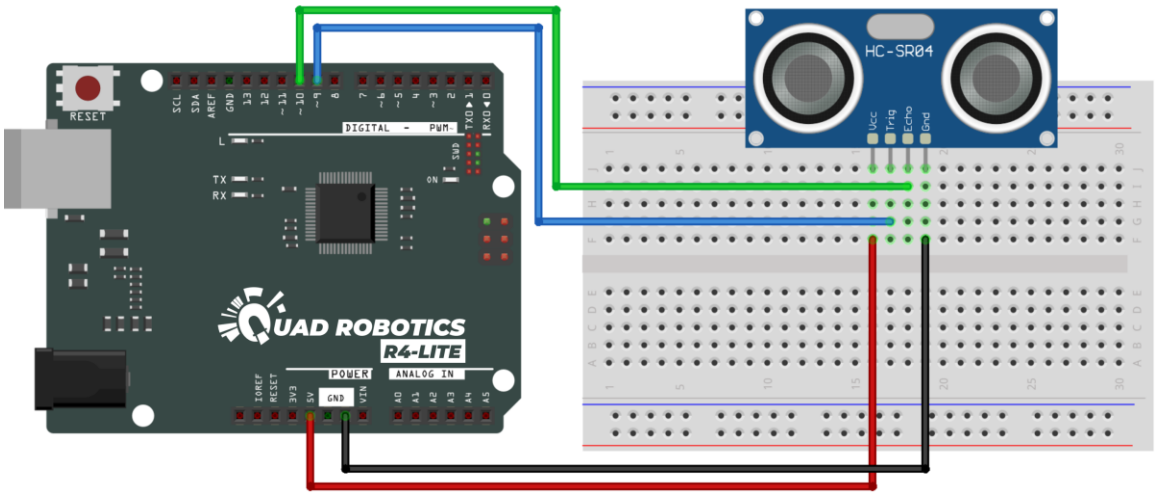
✨ Objective: Measure distance using HC-SR04 ultrasonic sensor.

Components Required

- UNO R4 board
- Ultrasonic Sensor
- Breadboard
- Jumper wires

Circuit Diagram

Trig → pin 9, Echo → pin 10, VCC → 5V, GND → GND.



Code

```
int trig = 9;
int echo = 10;
long duration;
int distance;

void setup() {
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);
  Serial.begin(9600);
}

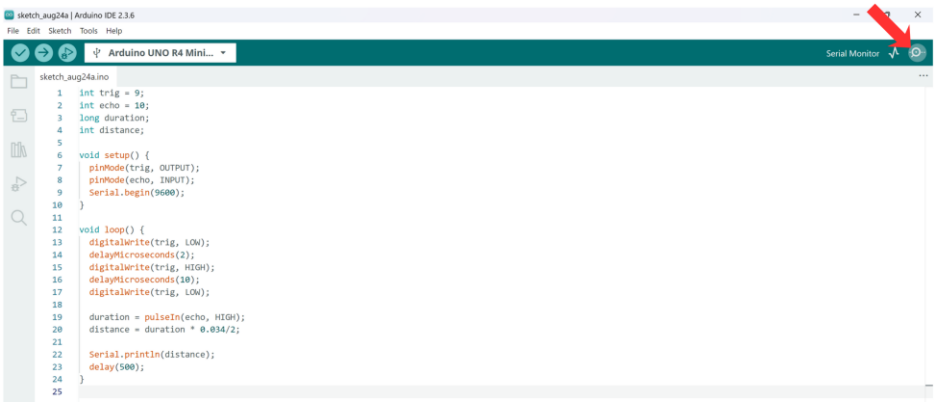
void loop() {
  digitalWrite(trig, LOW);
  delayMicroseconds(2);
  digitalWrite(trig, HIGH);
  delayMicroseconds(10);
  digitalWrite(trig, LOW);
```

```
  duration = pulseIn(echo, HIGH);
  distance = duration * 0.034/2;

  Serial.println(distance);
  delay(500);
}
```

Steps to Upload Code

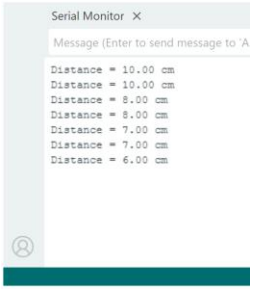
1. Connect your **UNO R4 board** to the computer using a **USB cable**.
2. Open **Arduino IDE** on your computer.
3. Go to **Tools > Board** and select **Arduino UNO R4**.
4. Go to **Tools > Port** and choose the correct **COM port**.
5. Either **copy and paste the above code** into the Arduino IDE OR open the file **Ultrasonic.ino** from the provided **CODE folder**.
6. Click on the **Upload button** (right arrow icon).
7. Wait for the code to **compile and upload**. The board will start running the program automatically.
8. Open serial monitor by clicking on the serial icon.



Explanation & Output

Sensor sends ultrasonic pulse and measures echo time.

Distance is calculated using speed of sound and it is displayed in the serial monitor.



DIY Extension

Make obstacle avoidance robots.

Project 11: DHT11 Weather Monitor

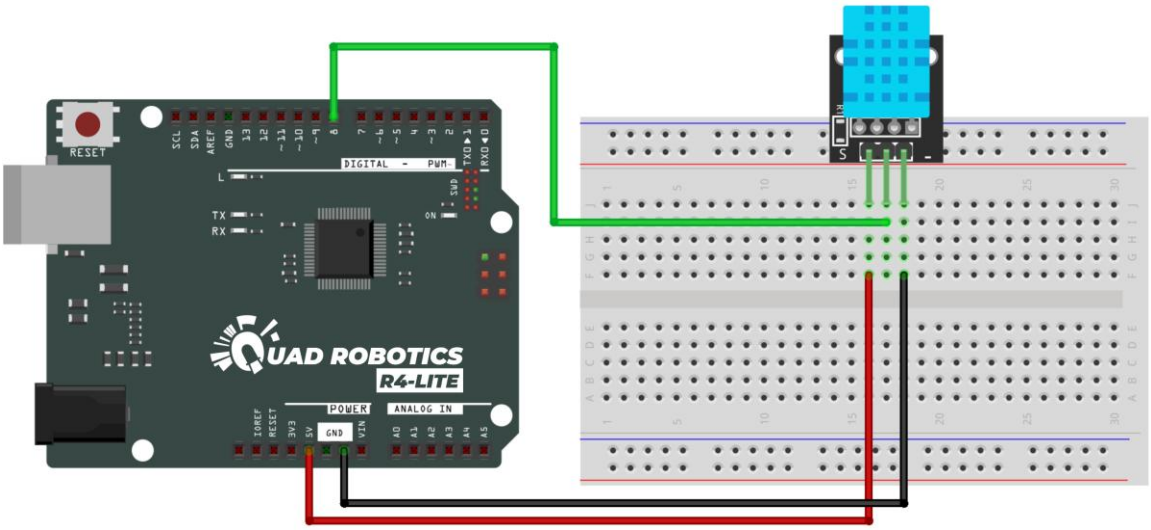
✨ Objective: Read temperature and humidity using DHT11 sensor.

Components Required

- UNO R4 board
- DHT11 Module
- Breadboard
- Jumper wires

Circuit Diagram

DHT11 data → pin 8, VCC → 5V, GND → GND.



Code

```
#include <DHT.h>
#define DHTPIN 8
#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(9600);
  dht.begin();
}

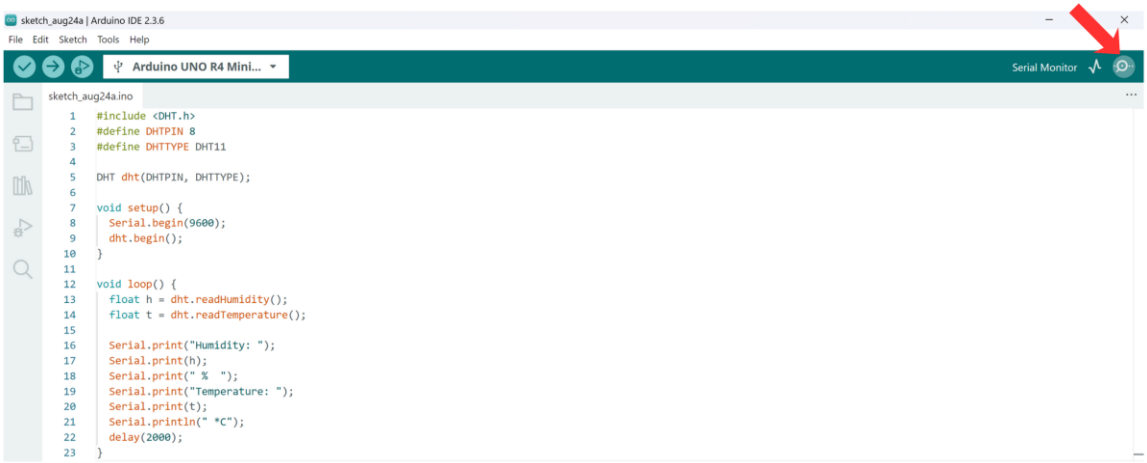
void loop() {
  float h = dht.readHumidity();
  float t = dht.readTemperature();

  Serial.print("Humidity: ");
  Serial.print(h);
  Serial.print(" %  ");
```

```
  Serial.print("Temperature: ");
  Serial.print(t);
  Serial.println(" *C");
  delay(2000);
}
```

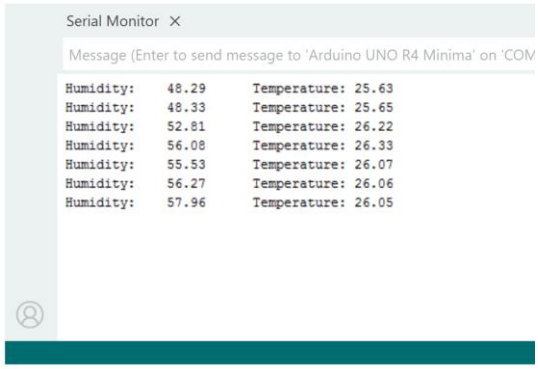
Steps to Upload Code

1. Connect your **UNO R4 board** to the computer using a **USB cable**.
2. Open **Arduino IDE** on your computer.
3. Go to **Tools > Board** and select **Arduino UNO R4**.
4. Go to **Tools > Port** and choose the correct **COM port**.
5. Either **copy and paste the above code** into the Arduino IDE OR open the file **DHT11Monitor.ino** from the provided **CODE folder**.
6. Click on the **Upload button** (right arrow icon).
7. Wait for the code to **compile and upload**. The board will start running the program automatically.
8. Open **serial monitor** by clicking on the serial icon.



Explanation & Output

DHT11 provides humidity and temperature readings.
Values are printed on Serial Monitor.



DIY Extension

Make your own weather station.

Project 12: Fire Sensor Alarm

✨ Objective: Detect fire and trigger buzzer.

Components Required

- UNO R4 board

- Fire Sensor Module

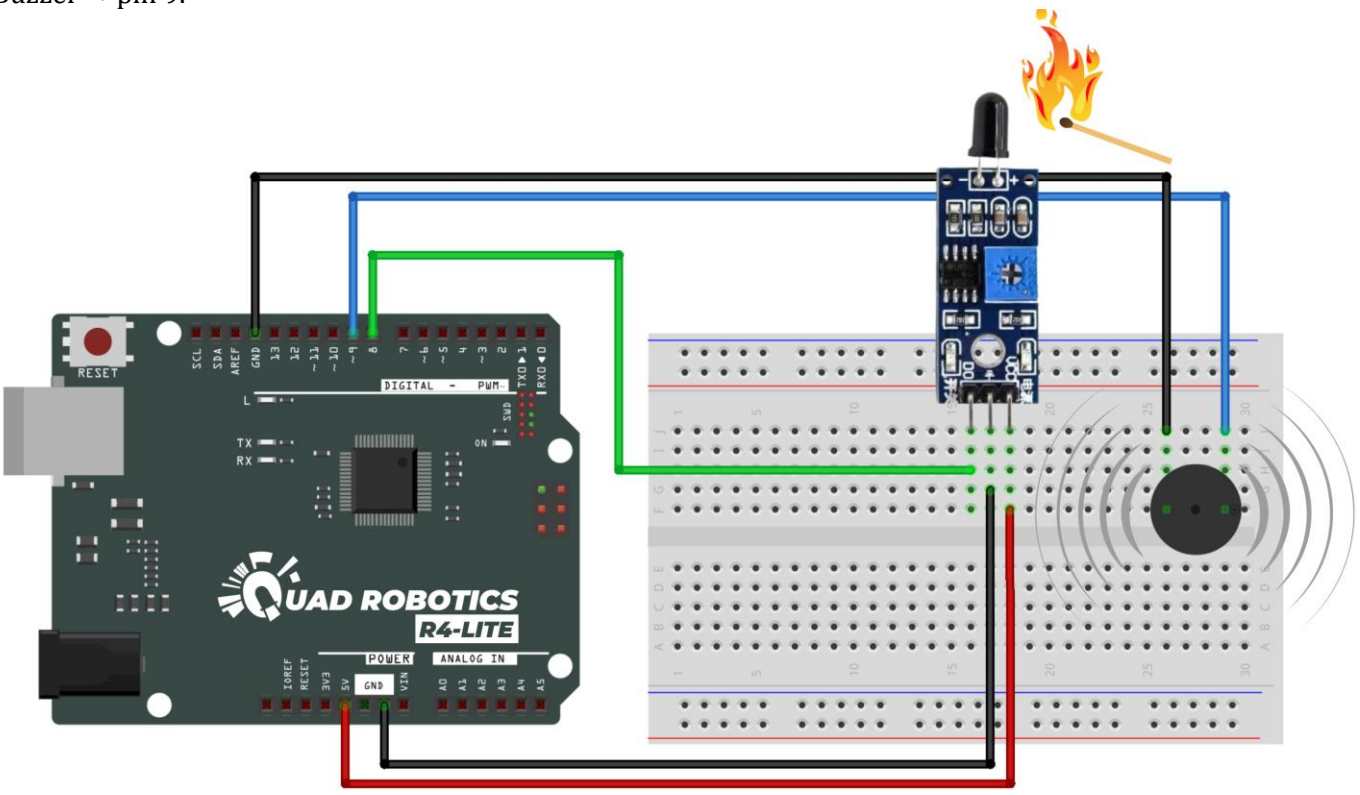
- Buzzer

- Breadboard

- Jumper wires

Circuit Diagram

Fire sensor output → pin 8,
Buzzer → pin 9.



💻 Code

```
int fire = 8;
int buzzer = 9;

void setup() {
  pinMode(fire, INPUT);
  pinMode(buzzer, OUTPUT);
}

void loop() {
  if(digitalRead(fire) == HIGH){
    digitalWrite(buzzer, HIGH);
  } else {
    digitalWrite(buzzer, LOW);
  }
}
```

📲 Steps to Upload Code

1. Connect your **UNO R4 board** to the computer using a **USB cable**.
2. Open **Arduino IDE** on your computer.
3. Go to **Tools > Board** and select **Arduino UNO R4**.
4. Go to **Tools > Port** and choose the correct **COM port**.
5. Either **copy and paste the above code** into the Arduino IDE OR open the file **FireAlarm.ino** from the provided **CODE folder**.
6. Click on the **Upload button** (right arrow icon).
7. Wait for the code to **compile and upload**. The board will start running the program automatically.

🔑 Explanation & Output

Fire sensor detects flames or strong IR light.

If fire detected, buzzer turns ON.

🛠 DIY Extension

Use as a safety alarm.

Project 13: Infrared Obstacle Detector

✨ Objective: Detect objects using IR sensor.

Components Required

- UNO R4 board

- IR Sensor Module

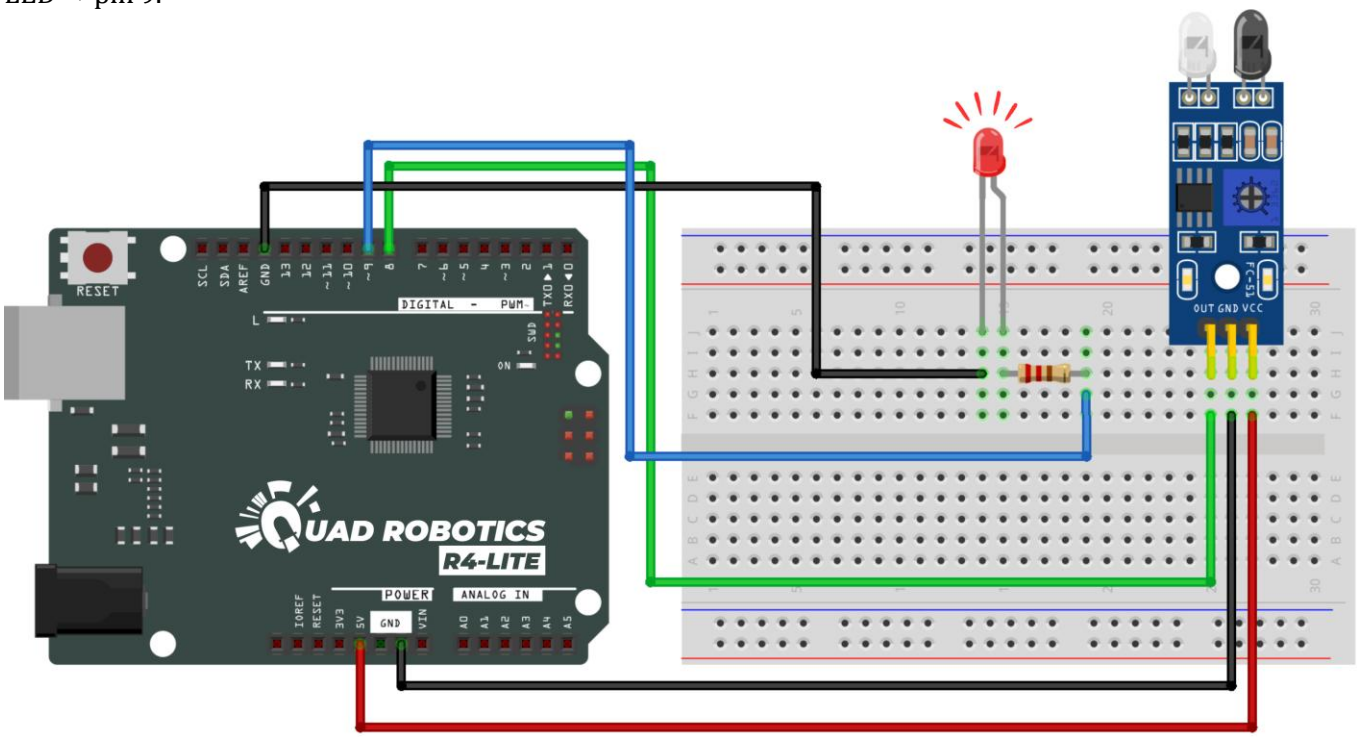
- LED or Buzzer

- Breadboard

- Jumper wires

Circuit Diagram

IR sensor output → pin 8,
LED → pin 9.



💻 Code

```
int ir = 8;
int led = 9;

void setup() {
  pinMode(ir, INPUT);
  pinMode(led, OUTPUT);
}

void loop() {
  if(digitalRead(ir) == LOW){
    digitalWrite(led, HIGH);
  } else {
    digitalWrite(led, LOW);
  }
}
```

📥 Steps to Upload Code

1. Connect your **UNO R4 board** to the computer using a **USB cable**.
2. Open **Arduino IDE** on your computer.
3. Go to **Tools > Board** and select **Arduino UNO R4**.
4. Go to **Tools > Port** and choose the correct **COM port**.
5. Either **copy and paste the above code** into the Arduino IDE OR open the file **IRDetector.ino** from the provided **CODE folder**.
6. Click on the **Upload button** (right arrow icon).
7. Wait for the code to **compile and upload**. The board will start running the program automatically.

🔑 Explanation & Output

IR sensor detects reflected infrared light from objects.

When obstacle is detected, it turns on the LED.

🛠️ DIY Extension

Use in line-following robots.

Project 14: OLED Display Basics

✨ Objective: Display text on 0.96 inch OLED.

Components Required

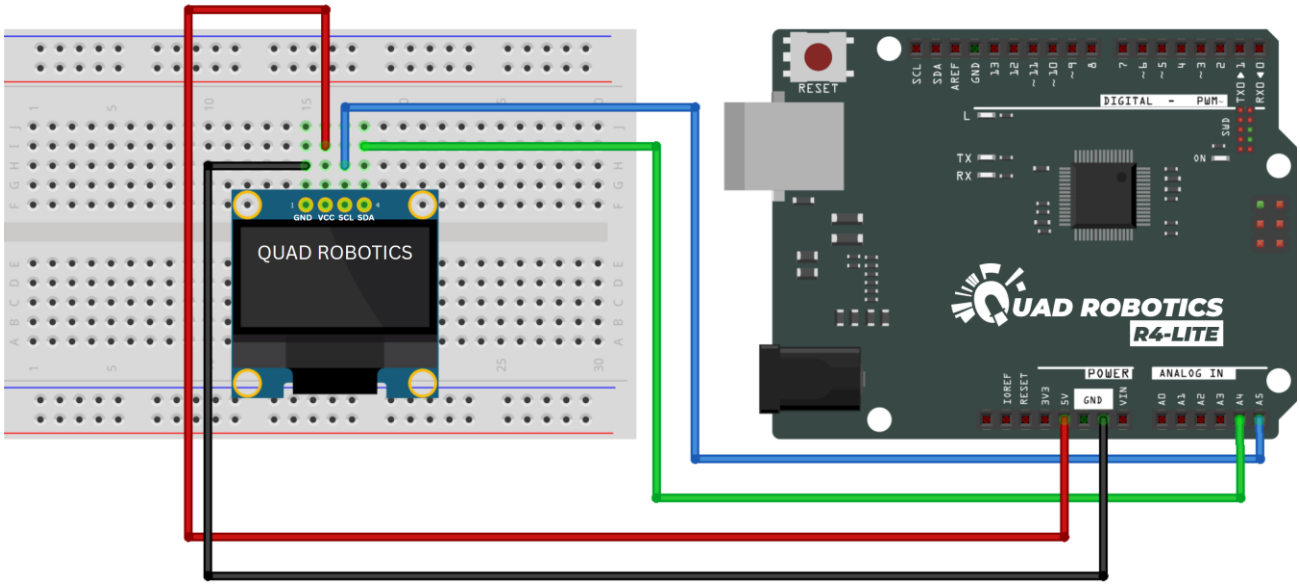
- UNO R4 board

- 0.96 inch OLED Display (I2C)

- Jumper wires

Circuit Diagram

VCC → 5V, GND → GND, SDA → A4, SCL → A5.



Steps to Upload Code

1. Connect your **UNO R4 board** to the computer using a **USB cable**.
2. Open **Arduino IDE** on your computer.
3. Go to **Tools > Board** and select **Arduino UNO R4**.
4. Go to **Tools > Port** and choose the correct **COM port**.
5. Either **copy and paste the below code** into the Arduino IDE OR open the file **OLEDDisplay.ino** from the provided **CODE folder**.
6. Click on the **Upload button** (right arrow icon).
7. Wait for the code to **compile and upload**. The board will start running the program automatically.

💻 Code

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

void setup() {
  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    for(;;);
  }
  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(SSD1306_WHITE);
  display.setCursor(0,10);
  display.println("QUAD ROBOTICS!");
  display.display();
}

void loop() {
}
```

🔑 Explanation & Output

The OLED uses I2C communication (SDA, SCL).

We display simple text “QUAD ROBOTICS” on screen.

🛠 DIY Extension

Show sensor values on OLED.

Make a mini weather station display.

📘 Closing Summary

Congratulations! 🎉 You have completed all 14 projects in this UNO R4 handbook.

By now, you have learned how to use LEDs, buzzers, sensors, motors, and displays with the UNO R4.

🚀 What's Next?

- Try combining multiple sensors and outputs to make smart projects.

- Create mini-robotics applications like obstacle avoiders, smart parking system.

- Build a small weather station using DHT11 + OLED display.

- Explore Arduino libraries and expand with IoT and wireless modules.

Keep experimenting and innovating! 🔧🤖